Data Science
SQL Exercise: Aggregate Functions and Grouping

1. COUNT() Function
Write a SQL query to find the total number of employees in the company.

```
SELECT COUN(*) AS Number_of_employees
FROM employees;
```

| Number_of_employees |
|---|
| 10 |

2. SUM() Function
Write a SQL query to find the total salary paid to all employees in the IT department.

```
SELECT SUM(salary) AS total_salary
FROM employees
WHERE department = 'IT';
```

| total_salary |
|---|
| 220000 |

3. AVG() Function
Write a SQL query to ~~find~~ calculate the average salary of employees in the HR department.

```
SELECT AVG(salary) AS average_salary
FROM employees
WHERE department = 'HR';
```

| average_salary |
| --- |
| 49 500 |

4. MIN() and MAX() Functions
Write a SQL query to find the highest and lowes salary in the company.

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary
FROM employees;
```

| Lowest_Salary | highest_Salary |
| --- | --- |
| 48 000 | 62 000 |

5. GROUP BY Statement
Write a SQL query to group employees by department and display the total salary paid in each department.

```
SELECT department,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department;
```

Result

| department | total_salary |
| --- | --- |
| Finance | 119 000 |
| HR | 99 000 |
| IT | 22 000 |
| Marketing | 105 000 |

6. GROUP BY and COUNT()
Write a SQL query to count how many employees work in each City.

```
SELECT city,
        COUNT(*) AS Number_of_employees
FROM employees
  GROUP BY City;
```

| City | Number_of_employees |
|---|---|
| Chicago | 3 |
| New York | 2 |
| Los Angeles | 2 |
| San Francisco | 2 |
| Houston | 1 |

7. GROUP BY and ORDER BY
Write a SQL query to group employees by department, calculate the average salary in each department, and order the result in descending order of average salary.

```
SELECT department, AVG(salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC;
```

| department | average_salary |
|---|---|
| IT | 55000 |
| Finance | 59500 |
| Marketing | 52500 |
| HR | 49500 |

8. HAVING Clause
Write a SQL query to find department where the total salary paid exceed 100 000.
(use GROUP BY and HAVING)

```
SELECT department, SUM(Salary) AS total_Salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100 000
```

| department | total_Salary |
|---|---|
| IT | 220 000 |
| Finance | 119 000 |

9. Combining GROUP BY, HAVING and ORDER BY
Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
SELECT city, COUNT(*) AS Number_of_employees
FROM employees
GROUP BY city
HAVING COUNT(*) > 1
ORDER BY Number_of_employees DESC;
```

| City | Number_of_employees |
|---|---|
| Chicago | 3 |
| New York | 2 |
| Los Angeles | 2 |
| San Francisco | 2 |

10. Combining Aggregate Functions
Write a SQL query to find the department with the highest average salary.

```
SELECT department,
       AVG(salary) As average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC
LIMIT 1;
```

| department | average_salary |
|---|---|
| Finance | 59500 |